A Two-Datum Obstruction to Representing Relative-Critical Sets by a Data-Independent Tikhonov Regularizer

Automatic math-discovery packet

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Source Conjecture

The source is Daniel Obmann and Markus Haltmeier, *Convergence analysis of critical point regularization with non-convex regularizers*, arXiv:2207.14480v2 [1]. In Section 4.3, pages 19–20, the authors ask whether their regularization by $\alpha\phi$-critical points can be equivalent to ordinary Tikhonov regularization with a modified penalty $\mathcal{R}_\phi$. They conjecture:

> "the proposed regularization is not equivalent to Tikhonov regularization independent of the choice $\mathcal{R}_\phi$."

The source discussion makes this a set-valued comparison: the full set of $\alpha\phi$-critical points is to coincide with the full minimizer set of a Tikhonov functional using one modified regularizer independent of the datum. The complete source passage spans the following two crops.

Definitions

For a function $F : X \to \mathbb{R}$ and a bound $\psi : X \to [0, \infty)$, the source defines $r \in \partial_\psi F(x)$ by

$$F(x) + \langle r, u - x\rangle \le F(u) + \psi(u) \qquad (u \in X).$$

The point x is ψ-critical if $0 \in \partial_\psi F(x)$. Consequently,

$$x \in \operatorname{crit}_\psi F \quad \iff \quad F(x) \le \inf_{u\in X} \big(F(u) + \psi(u)\big). \tag{1}$$

This is also Lemma 2.4(6) of the source paper.

For data $y \in Y$, a linear forward map $K : X \to Y$, a penalty $\mathcal{R}$, and $\alpha > 0$, the Tikhonov functional is

$$\mathcal{T}_{\alpha,y}(x) = \frac{1}{2}\|Kx - y\|^2 + \alpha\mathcal{R}(x).$$

An ordinary modified-penalty representation of its relative-critical sets would be a proper function $S : X \to (-\infty, +\infty]$, independent of y, such that

$$\operatorname{crit}_{\alpha\phi} \mathcal{T}_{\alpha,y} = \arg\min_x \left\{\frac{1}{2}\|Kx - y\|^2 + \alpha S(x)\right\}$$

for all data under consideration.

4.3 Non-equivalence to Tikhonov regularization

One might conjecture that the proposed regularization with $\alpha\phi$-critical points of $\mathcal{T}_{\alpha,y_\delta}$ is equivalent to Tikhonov regularization for some other modified choice of the regularizer. While we cannot give a definite answer to this question at this point, we conjecture that this is not the case.

To support our hypothesis, let us analyze what would happen if the construction of $\alpha\phi$-critical points of $\mathcal{T}_{\alpha,\delta}$ were equivalent to the Tikhonov regularization with some regularizer $\mathcal{R}_\phi$. In this

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case, the limiting problems would also coincide and thus for any $y \in \operatorname{ran}(\mathbf{K})$ and $x \in \mathbb{X}$ we have

$$\mathcal{R}(x) \leq \min_{\mathbf{K}u=y} \mathcal{R}(u) + \phi(u) \iff \mathcal{R}_\phi(x) \leq \min_{\mathbf{K}u=y} \mathcal{R}_\phi(u) \,.$$

Denoting by x_ϕ a solution of $\min_{\mathbf{K}u=y} \mathcal{R}(u) + \phi(u)$ we have $\mathcal{R}(x) \leq \mathcal{R}(x_\phi) + \phi(x_\phi)$. Clearly, this is the case if and only if $\max\{\mathcal{R}, \mathcal{R}(x_\phi) + \phi(x_\phi)\}$ is minimal among all possible solutions. This essentially means that $\mathcal{R}_\phi = \max\{\mathcal{R}, \mathcal{R}(x_\phi) + \phi(x_\phi)\}$.

While such a choice can theoretically be used to characterize the limiting problem, we do not have access to x_ϕ and therefore cannot work with $\mathcal{R}_\phi$ in practice. A slightly more subtle problem is that $\mathcal{R}_\phi$ depends on the exact data and as such cannot be used for the case of noisy case where $\alpha > 0$. Thus, we conjecture that the proposed regularization is not equivalent to Tikhonov regularization independent of the choice $\mathcal{R}_\phi$.

4.4 ReLU-Networks as class of possible regularizers

Figure 1: Section 4.3 of the source paper, pages 19–20.

Proof Intuition

It is enough to use the most elementary admissible instance of the source framework: one dimension, identity forward map, zero original regularizer, and a positive constant relative-error bound. Formula (1) then turns the relative-critical set into a closed interval of fixed positive radius centered at the datum.

Two nearby data give two overlapping intervals. Choose their two centers so that both centers lie in both intervals. If one data-independent penalty made both intervals minimizer sets, then the modified Tikhonov objective would have equal values at the two centers for each datum. Moving the quadratic data term from one center to the other reverses its sign, forcing the same penalty difference to equal both ε and $-\varepsilon$. This is the desired obstruction.

Main Theorem

Theorem 1 (No data-independent modified penalty)**.** *Fix $\alpha > 0$ and $\varepsilon > 0$. Let*

$$X = Y = \mathbb{R}, \qquad K = I, \qquad \mathcal{R} \equiv 0, \qquad \phi \equiv \varepsilon,$$

and put $\rho = \sqrt{2\alpha\varepsilon}$. Then for every $y \in \mathbb{R}$,

$$\operatorname{crit}_{\alpha\phi} \mathcal{T}_{\alpha,y} = [y - \rho, y + \rho].$$

There is no proper function $S : \mathbb{R} \to (-\infty, +\infty]$ for which

$$\operatorname*{arg\,min}_{x \in \mathbb{R}} \left\{ \frac{1}{2}(x - y)^2 + \alpha S(x) \right\} = \operatorname{crit}_{\alpha\phi} \mathcal{T}_{\alpha,y} \tag{2}$$

holds for every $y \in \mathbb{R}$. In fact, no such S can make (2) hold simultaneously for the two data values $y = 0$ and $y = \rho$.

Proof. Here

$$\mathcal{T}_{\alpha,y}(x) = \frac{1}{2}(x - y)^2 \quad \text{and} \quad \inf_u \big(\mathcal{T}_{\alpha,y}(u) + \alpha\phi(u)\big) = \alpha\varepsilon.$$

Therefore (1) gives

$$x \in \operatorname{crit}_{\alpha\phi} \mathcal{T}_{\alpha,y} \quad \Longleftrightarrow \quad \frac{1}{2}(x - y)^2 \le \alpha\varepsilon \quad \Longleftrightarrow \quad x \in [y - \rho, y + \rho].$$

Suppose that a proper S represents the relative-critical sets for $y = 0$ and $y = \rho$. Both points 0 and ρ belong to both relevant intervals:

$$0, \rho \in [-\rho, \rho] \quad \text{and} \quad 0, \rho \in [0, 2\rho].$$

Hence both are global minimizers of the modified Tikhonov objective for each of the two data. Since S is proper, the corresponding minimum values are finite. Equality of the values at 0 and ρ for datum $y = 0$ gives

$$\alpha S(0) = \frac{1}{2}\rho^2 + \alpha S(\rho), \qquad \text{so} \qquad S(0) - S(\rho) = \frac{\rho^2}{2\alpha} = \varepsilon.$$

For datum $y = \rho$, equality at the same two minimizers gives

$$\frac{1}{2}\rho^2 + \alpha S(0) = \alpha S(\rho), \qquad \text{so} \qquad S(\rho) - S(0) = \varepsilon.$$

Adding the last two identities yields $0 = 2\varepsilon$, contradicting $\varepsilon > 0$. Thus no data-independent modified penalty exists. □

Remark 1 (Strength of the obstruction)**.** *No convexity, continuity, lower semicontinuity, or coercivity is assumed of S. The contradiction also fixes α, so allowing the modified penalty to depend on α, on ε, or on the original regularizer does not help. Dependence on the datum is the only excluded dependence, exactly as in the source conjecture.*

Verification Notes

The proof uses only the definition of relative criticality and equality of an objective at two global minimizers. The accompanying script `code/verify_two_data_obstruction.py` checks, for several positive (α, ε) pairs, that the two selected points lie in both relative-critical intervals and that the two minimizer equalities demand opposite penalty differences ε and $-\varepsilon$. Running

```
python code/verify_two_data_obstruction.py
```

prints `all checks passed`. These finite checks are only a sanity test; the symbolic argument above proves the theorem for every $\alpha, \varepsilon > 0$.

Limitations and Novelty Check

The theorem settles the set-valued comparison made in Section 4.3: it shows that critical-point regularization cannot *in general* be recast as ordinary Tikhonov minimization with a data-independent penalty. It does not claim that every single-valued algorithmic selection from a relative-critical set is non-Tikhonov. In this example, for instance, the selection $x(y) = y$ is represented by the zero penalty even though the entire interval-valued map is not.

The run's lightweight registry, solution, attempt, and proof-gap indexes were searched for arXiv:2207.14480 and the core equivalence terminology. Exact queries of the official arXiv API on August 11, 2026 for "critical point regularization" and "relative sub-differentiability" found the source, the authors' later convergence-rates paper [2], and an unrelated paper on another relative-subdifferential notion. The later follow-up does not discuss or settle the equivalence conjecture. No arXiv answer was found within these bounded exact-query searches.

Human Review Recommendation

Verify that the source's word "equivalent" is intended in the set-valued sense made explicit by its displayed biconditional and its discussion of the whole set of $\alpha\phi$-critical points. Under that interpretation, the example satisfies the paper's standing regularization assumptions: $\mathcal{R} = 0$ is nonnegative and weakly lower semicontinuous, $\mathcal{T}_{\alpha,y}$ is coercive because $K = I$, and the constant $\phi = \varepsilon$ is an admissible relative bound.

References

[1] D. Obmann and M. Haltmeier, *Convergence analysis of critical point regularization with non-convex regularizers*, Inverse Problems 39 (2023), 095002; arXiv:2207.14480v2.

[2] D. Obmann, *Convergence rates for critical point regularization*, arXiv:2302.08830 (2023).